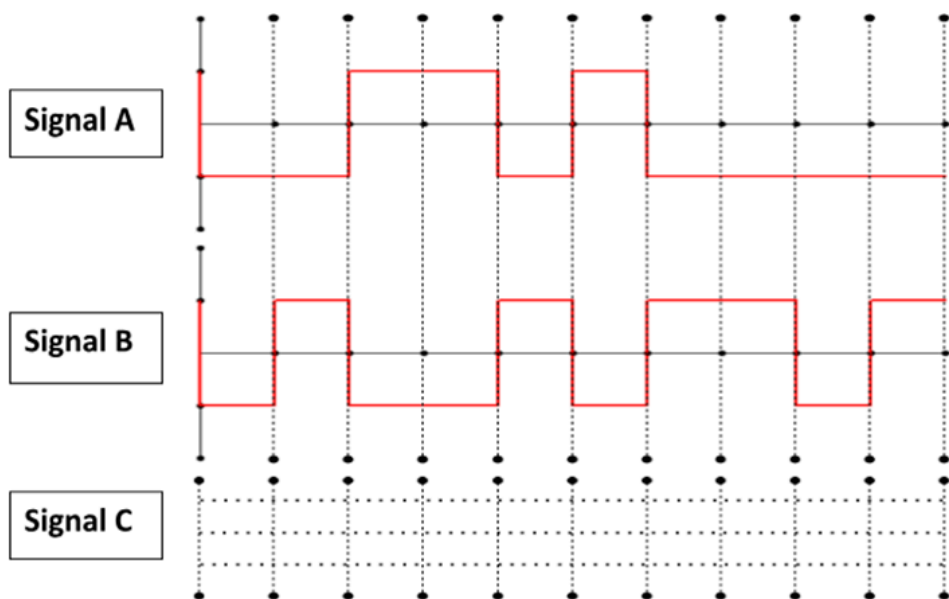
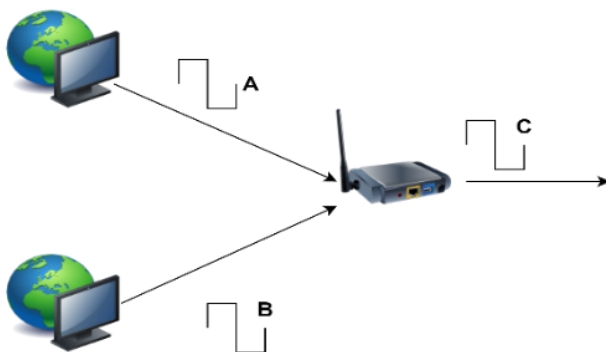


CSE320 (Data Comm) – Practice Sheet on Chapter 4

Q1

Two devices A and B are sending digital signals using the NRZ-I line coding scheme. Device C is receiving the signal simultaneously and combining them using bitwise AND operation. Then produces the final digital signal using a line coding scheme that doesn't have the consecutive 0 problem. Illustrate the final signal produced by C. [You can use any valid line coding scheme for C] [Draw it in the question paper only.]

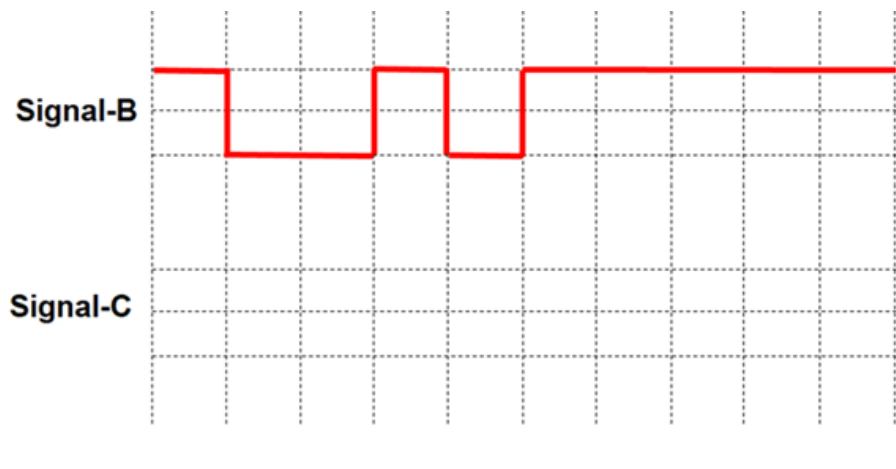


Q2

Construct a mapping table for a 3B/4B Block Coder with the following rules. You can ignore the control sequences, just construct the table for encoded sequences. While choosing the encoded sequences, make sure to follow the rules specified below:

- The encoded sequences can't have three consecutive zeros (0) in the table
- If first bit is 1, it has to be followed by a 0 (i.e. You can consider encoded sequences that start with "10")
- If the first bit is 1, it cannot be followed by a 1 (i.e. Ignore encoded sequences that start with "11")
- "0111" is reserved for the control sequence and can't be used in the mapping.

Q3



- Decode the Signal B and generate the binary bit stream using NRZ-I line coding scheme.
- Now from the extracted binary bits, draw the Signal-C where there is no synchronization problem.

Q4

Convert the following bit stream to a signal using an appropriate encoding scheme that matches the requirements given. Write which signal encoding scheme you are using.

Data: 1 0 1 0 1 1 0 1 1 1 0 0 0 0 1

Requirements

Scenario 1: The encoding scheme must occupy a low bandwidth. Any self-synchronization and DC component problem may be ignored.

Scenario 2: The encoding scheme must be self-synchronizing and should not have a DC component problem. High bandwidth is acceptable.

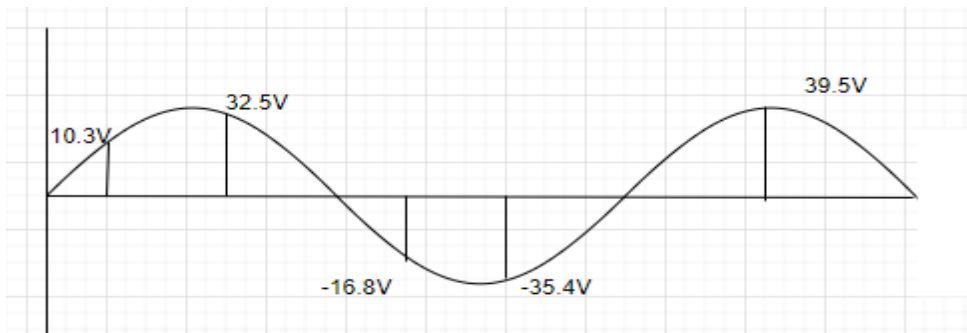
Q5

The following table depicts a sampled analog signal for digital signal representation. By applying the concept of Pulse Code Modulation, assume there will be 3-bit code words for each sampled amplitude. Show the normalized quantized value and quantization code for the given analog signal value at different time stamps. Assume that the sampling amplitudes are between -20V to +20V.

Time	Analog Signal Value (V)
0	-3.3
1	9.4
2	15.7
3	-13.2

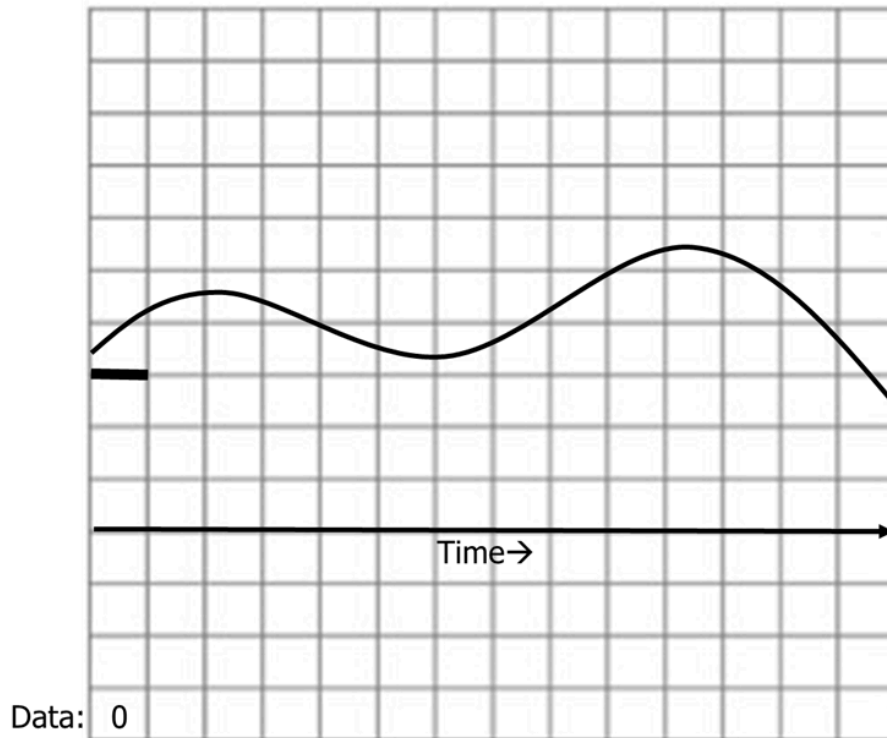
Q6

The following figure depicts a sampled analog signal for digital signal representation. By applying the concept of Pulse Code Modulation, assume there will be 3-bit code words for each sampled amplitude. Show the normalized Quantized value and quantization code for the given analog signal value at different time stamps. Assume that the sampling amplitudes are between -40V to +40V.



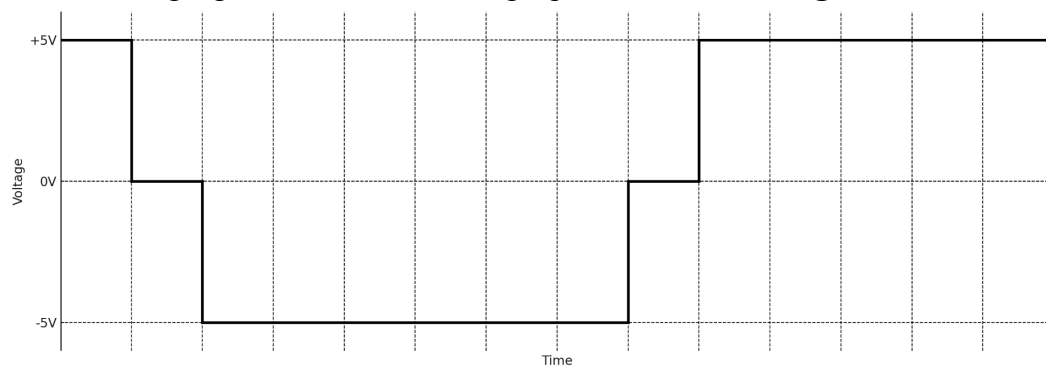
Q7

Show the staircase in the following graph and generate the digital data from the given analog signal using the Delta Modulation (DM) technique.



Q8

The following signal was obtained using a particular **line coding** scheme.



- I. Deduce which line coding scheme was used and decode the original bit stream.
- II. Explain whether this signal suffers from baseline wandering and DC components issue.
- III. Redraw the signal using a suitable scrambling technique that does not suffer from these drawbacks of consecutive 4 zeros.

